

Kushar Dogra

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PROFESSIONAL SUMMARY

Cybersecurity graduate (MS, Northeastern University, May 2026) with hands-on experience across penetration testing, application security, threat detection, SOC operations, vulnerability management, and GRC compliance. Background as a Full Stack Developer provides a unique developer-security perspective — skilled in Python, PowerShell, and Bash for security automation, and experienced building production systems with security-by-design principles. CompTIA Security+ certified (August 2026) and holds an active HackerOne bug bounty account. Seeking a full-time cybersecurity role where development experience and security depth combine to deliver measurable impact.

TECHNICAL SKILLS

Languages & Frameworks: Python, Django, Node.js, JavaScript, PowerShell, Bash, SQL, Flask, REST APIs, Go (learning)

Security Tools: Burp Suite, Metasploit, Nmap, Gobuster, SQLMap, Snort IDS/IPS, Splunk, Wireshark, Ghidra, GDB, pwntools, aircrack-ng, John the Ripper, tcpdump, ntopng

Frameworks & Compliance: OWASP Top 10, MITRE ATT&CK, NIST 800-53, NIST CSF, ISO 27001, CIS Benchmarks, HIPAA, PCI-DSS, GDPR, SOC 2, CVSS, Cyber Kill Chain

Cloud & DevOps: AWS EC2, Docker, Kubernetes, Git, CI/CD security awareness

Platforms: Kali Linux, Ubuntu, Windows

WORK EXPERIENCE

Application Security-Focused Full Stack Developer

Aug 2023 – Aug 2024

Quindl Technologies | Coimbatore, India

- Built 4 production web applications (e-commerce, booking system, analytics dashboards) using Python, Django, Node.js, JavaScript, MySQL, and SQLite — delivering secure-by-design systems across the full SDLC within a 12-month engineering lifecycle
- Identified insecure coding patterns and implemented Django security hardening — CSRF token enforcement, input validation, secure session management, and HTTPS enforcement — reducing the application attack surface across all 4 deployed apps
- Designed database schemas with role-based access controls (RBAC) and least-privilege principles — preventing unauthorized data access and enforcing access control across all user account types
- Led security-focused code reviews across a 15-person developer team, establishing secure coding standards and eliminating vulnerable patterns — resulting in consistent security hygiene and zero untested deployments across all team commits

CYBERSECURITY PROJECTS

Capstone: EPAC Compliance Automation Framework

Jan 2026 – Apr 2026

- Architected and developed EPAC, a cybersecurity automation framework using Python, Flask, PowerShell, and JavaScript that automates 21 security controls across 5 domains for Windows endpoints — achieving a measured 95% auto-fix rate for misconfigurations and reducing multi-framework audit preparation effort by 70%
- Designed compliance mapping architecture aligning controls to NIST 800-53, HIPAA, and CIS Benchmarks, with MITRE ATT&CK technique mapping for each control domain — built RESTful API with real-time SSE streaming and interactive security dashboard for continuous endpoint monitoring
- Integrated detection logic into Sentrinus Endpoint Protection Platform (Cors Lab Technology) — contributing MITRE ATT&CK-mapped detection rules and compliance automation currently in development and testing phase

Full Penetration Testing Engagement

- Executed a complete 5-phase penetration test against an Ubuntu 24.04 LTS target — performed network reconnaissance (Nmap across all 65,535 ports identifying 7 services), web enumeration (Gobuster uncovering exposed .git repository with deleted credentials), and credential recovery (zip2john + John the Ripper) — achieving lateral movement and privilege escalation to root shell (uid=0, gid=0) via Samba CVE exploitation (Metasploit exploit/linux/samba/is_known_pipename)
- Documented complete kill chain with MITRE ATT&CK mapping at each stage, CVSS-scored findings, reproduction steps, and prioritized remediation recommendations — producing an audit-ready penetration test report covering all 4 captured flags and full system compromise

Snort IDS/IPS Threat Detection Lab

- Authored 10+ custom Snort detection rules targeting SQL injection, three CVE-based attack signatures (CVE-2021-26855 Exchange SSRF, CVE-2022-1388 F5 BIG-IP auth bypass, CVE-2018-7600 Drupalgeddon2 RCE), and data exfiltration patterns — integrated with Splunk SIEM for centralized log correlation and achieved 100% true positive rate with zero false negatives

- Deployed AWS EC2 honeypot monitored via ntopng/Netflow — captured and triaged 1,024 live network flows from real-world attack campaigns, escalating 15 critical alerts from RDP, SMB, SSH, and MySQL brute-force attempts mapped to MITRE ATT&CK T1021.001, T1571, and T1046
- Performed PCAP malware analysis in Wireshark — identified C2 beaconing to Moldova-based infrastructure (89.187.51.0), extracted IOCs (load.exe 19KB, ribbon.tar), and produced structured incident report with containment and remediation recommendations

Web Application Security Testing

- Conducted manual and automated web application security assessment — exploited SQL injection (OR 1=1 auth bypass via SQLMap and Burp Suite), command injection (;cat /etc/passwd achieving RCE with SSH key extraction), and broken access control vulnerabilities, producing CVSS-scored findings with code-level remediation guidance for each vulnerability class
- Configured Burp Suite as HTTPS man-in-the-middle proxy to assess TLS implementation — intercepted and decrypted encrypted traffic, performed session token analysis and replay attacks, demonstrating that transport encryption alone provides zero protection against application-layer injection attacks

OSINT & Reconnaissance Research

- Conducted passive threat intelligence gathering using Spiderfoot — collected 1,500+ data elements across 47 data types including 25+ exposed corporate email addresses, 12 web servers, and 8 high-risk correlation findings from public sources without touching target systems
- Performed internet-wide attack surface mapping using Shodan and Censys — identified exposed IoT devices, CVE-affected services, and infrastructure leaked via SSL certificates; followed by Amass subdomain enumeration discovering 50+ subdomains and 5 exposed admin panels

Security Architecture & Network Fundamentals

- Designed a 10-zone enterprise network segmentation architecture for 5,000+ users — implemented CIDR subnetting across Finance, HR, IT, Legal, Sales, DMZ, Database, VPN, Guest WiFi, and Admin zones with Splunk SIEM, Symantec DLP, and FortiWeb WAF positioned across trust boundaries
- Threat-modeled the architecture to identify 3 critical attack paths — lateral movement via Guest-DMZ pivot, credential theft through VPN-DB trust, and PII exfiltration via HR-Finance relationship — documented mitigating controls mapped to NIST CSF and MITRE ATT&CK

Wireless Security & Protocol Analysis

- Assessed WPA2 security posture by capturing EAPOL 4-way handshake and performing offline dictionary attack with aircrack-ng (140,256 candidates at 2,847 keys/sec) — demonstrated critical risk of weak passphrases and produced remediation recommendations covering passphrase policy and network segmentation
- Implemented S/MIME end-to-end email encryption with X.509 certificate from Actalis CA — configured AES-256 encryption and RSA digital signatures in Thunderbird, verified authenticity via SHA-256 fingerprint comparison over out-of-band channel; analyzed PCAP forensics to reconstruct full attack timeline including C2 beaconing (TCP 4444, ~60s intervals) and cleartext FTP credential extraction

Binary Exploitation & Buffer Overflow Research

- Developed working exploits for stack-based buffer overflow vulnerabilities — reverse-engineered target binaries using Ghidra and objdump, calculated return address offsets via GDB, and constructed shellcode injection, ret2libc, and stack canary bypass exploit chains using pwntools — achieving root shell (SUID → uid=0) across all targets with a 96.5% CTF score

Security Risk Management Assessment

- Conducted end-to-end security risk assessment for two simulated organizations (federal payroll agency and private enterprise) — identified 10 assets, 11 threats, 11 vulnerabilities, and 12 threat/vulnerability pairs using FIPS 199 methodology, producing executive-level risk findings with ALE/SLE/ARO quantification across a 16-sheet Excel model
- Performed compliance gap analysis mapping 35 controls and 14 CISO policies across NIST 800-53, SOC 2, HIPAA, PCI-DSS, GDPR, and CCPA — achieving full coverage across all 15 MOT control families and closing 5 critical gaps including Cryptography, Identification & Authentication, and Security Training

CERTIFICATIONS & TRAINING

CompTIA Security+ – Earned: August 2026

TryHackMe — SOC Level 1 Path – In Progress

TryHackMe — Digital Forensics & Incident Response Path – In Progress

HackerOne Bug Bounty Platform – Active (Feb 2026 – Present)

EDUCATION

Northeastern University | Boston, MA

Sep 2024 – May 2026

Master of Science, Cybersecurity — Khoury College of Computer Sciences

Kumaraguru College of Technology | Coimbatore, India

Aug 2020 – May 2024

Bachelor of Technology, Information Technology